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6th sem

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
COE/B.TECH./CSE/PEC-CS601A/2024-25
2025
DATA WAREHOUSING & DATA MINING

Full Marks: 70 Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are requested to write their answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer all questions
1. Define data warehousing.
2. Define star schema of data warehousing.
3. Define dimension and fact table.
4. Draw a probable concept hierarchy for "location" dimension.
5. What is data mart?

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15=60

Answer the Question 6 and any ***three*** from the remaining questions

6. i) Define Entropy and information gain. State the criteria to identify a splitting attribute for decision tree. 3
   ii) What is the difference between clustering and classification on the basis of training dataset? 2
   iii) Draw a decision tree from the training data set given in Table 1 using information gain. {***cheat*** is the class label attribute}. List the classification rules derived from the decision tree. 10

TID	Marital Status	Taxable Income	Refund	Cheat			
1	Single	High	No	Yes	N	H	S
2	Married	High	No	No	N	H	M
3	Single	Low	No	No	N	L	S
4	Married	High	Yes	No	Y	H	M
5	Divorced	Average	No	Yes	N	A	D
6	Married	Low	No	No	N	L	M
7	Divorced	High	Yes	No	Y	H	D
8	Single	Average	No	Yes	N	A	S
9	Married	Low	No	No	N	L	M
10	Single	High	Yes	No	Y	H	S

Table 1

7. i) Draw the 3-tier architecture of data warehouse and state the functions of OLAP servers exist in the middle tier of the data warehouse architecture. 5
   ii) Write the differences between OLAP and OLTP. 5
   iii) Write short note on any one a) Snowflake schema b) k-means algorithm 5

PTO

8. i) State the apriori property and its role in the joining and pruning steps of apriori algorithm.

ii) Consider the given database (Table-1), minimum support count S=2. a) List all the candidate set and large frequent item-sets using Apriori Algorithm. B) Find 4 strong association rules for minimum confidence 70%. 5+3 =8

iii) If a system needs t seconds for scanning the dataset one time, determine the total number of scan operations and scanning time to compute the largest frequent item set. 3

Table-1

TranID	List of Item IDs
T100	Butter, Milk, Rice
T200	Bread, Butter, Jam
T300	Butter, Sugar
T400	Bread, Butter, Milk
T500	Bread, Sugar, Pepsi
T600	Butter, Sugar, Curd
T700	Bread, Sugar
T800	Bread, Butter, Sugar, Jam
T900	Bread, Butter, Sugar

3

9. i) Draw a lattice of cuboid for four dimensional (item, location, time, customer) data and show the base cuboid and apex cuboid. 3

ii) Construct a FP-Tree and draw all the conditional FP-Tree assuming minimum support count = 3 of the given dataset (Table-2). 5+4

iii) Compare apriori algorithm with FP-Tree approach for frequent itemset detection. 3

Table-2

Tran_ Id	List of items
T1	F, A, C, D, G, I, M, P
T2	A, B, C, F, L, M, O
T3	B, F, H, J, O
T4	B, C, K, S, P
T5	A, F, C, E, L, P, M, N

10. Suppose that a data warehouse for big-bazar consists of the four dimensions customer, city, product, and time, and two measures count and sales-amount. At the lowest conceptual level (i.e., for a given customer, city, product and time combination), the sales-amount measure stores the actual purchase amount of the customer.

At higher conceptual levels, sales-amount stores the total purchase amount for the given combination.

a) Draw a schema diagram for modeling the above data warehouse. State clearly the tables, facts & keys. 10

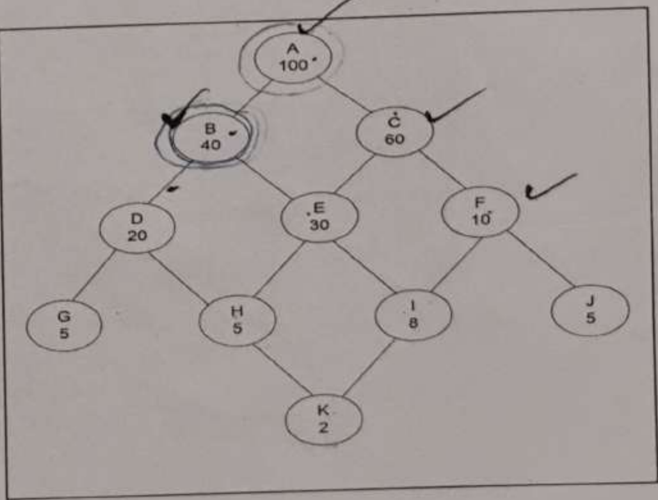
b) Starting with the base cuboid [customer, city, product and time], what specific OLAP operations should you perform in order to list the total sales amount of "product=Printer" for each country. 5

11. i) Consider the following lattice of views (Fig. 1) along with a representation of the number of rows in each view where A is the base cuboid. Consider that view A is already materialized. Find another three views for materialization from B-K views which provide maximum benefit. 9

ii) Show the cost optimality for the above view materialization. 3

iii) What is materialized view? What is the advantage of materialized view over non-materialized view? 3

Fig. 1



E

$C = 40 \times 7 = 280$

$D = 20 \times 4 = 80$

E =

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE

[A GOVERNMENT AUTONOMOUS COLLEGE]

JGEC/B.TECH./CSE /PCC-CS602/2024-25

2025

COMPUTER NETWORK

Full Marks: 70 Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer ***all*** questions 5x2=10

1. What is piggybacking?
2. Why is error control mechanism required at data link layer?
3. What is routing?
4. What is port address?
5. What are Net Id and Host Id in IP addressing scheme?

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any ***four*** questions 4x15=60

6. i) What is framing? What is bit stuffing and byte stuffing? 1+4=5
   ii) Define different types of class-full addressing scheme in Data Communication? 5
   iii) Given a 10 bits sequence 1010011110 and a divisor 1011. Find the CRC. Check your answer. 5

7. i) "In Selective-Repeat ARQ, sender window size cannot be >2^{m-1}". Justify it with a suitable pictorial example. 5
   ii) What is the advantage of two dimensional parity over simple parity? Explain with suitable example. 5
   iii) Suppose the following block of 16 bits (10101001 00111101) is to be sent using a checksum of 8 bits. Find Checksum at sender side and also check at receiver side to see whether there is an error or not. 5

8. i) Briefly describe key duties and responsibilities of different layers of TCP/IP model. 5
   ii) Suppose a organization is given the block 17.12.40.81/26. Now the organization has 3 offices and needs to divide the addresses into 3 sub-blocks of 16, 16, 32. Find the first address, last address and allotted IP (in x.t.z.t/n notation) for each block. 5
   iii) Why the size of the sliding window is n-1 instead of n? Explain with an example. Where, n is the sequence number to represent frame. 5

9. i) What are the different components of data communication? 3
   ii) Draw the header format of IPv4 datagram and give details of different segment. 5
   iii) Encode the data word 1011011 using Hamming code with even parity. Find the minimum number of parity/redundant bit and construct the Hamming code word. 7

10. Write the differences between 9.41 5x3=15
    i) TCP and UDP
    ii) Classless and Class-full addressing scheme 9.71
    iii) Logical address and Physical address 9. 9.12
    iv) Star and Mesh topology.
    v) IPV4 and IPV6 9.6 9.81 9.57

11. Write short notes on 3x5=15
    i) ARP
    ii) Three-way-handshaking with diagram
    iii) Go-Back-N ARQ protocol

END

525 8 9.71

$\frac{207+201}{21+21}$ $\frac{408}{42}$

product
price
place
promotion

analysis
and
decision

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JGEC/B.TECH/CSE/ HU601/2024-25

PRINCIPLES OF MANAGEMENT

Full Marks: 70 Time: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions: 5x2=10

1. What is administrative management? 2
2. What is the marketing information system? 2
3. What is the marketing mix? 2
4. What is a code of ethics? 2
5. What is stress? 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions: 4x15=60

6. Discuss various techniques of group decision making. Write a short essay on common errors in group decision-making. What are the characteristics of an effective marketing mix? 5+5+5

7. Write an essay on the elements of a marketing information system. What are the guidelines for effective decision-making? What are the ten commandments for effective leadership? 5+3+7

8. Examine why organization needs to build and maintain high ethical standards. How do training and development programmes help the organizations and their employees? 7+8

9. What is bureaucratic management? Write a note on its advantages and disadvantages. Make a difference between leaders and managers. 3+6+6

10. Write a note on trait theory of leadership. What are the fourteen principles of administration? Give an argument in favour of and against corporate social responsibility. 5+5+5

11. What are the sources of a leader's power? Write a note on the barriers to effective planning. Discuss various steps involved in the process of selecting a candidate for a job. 3+6+6

Delphi
Fish
Didactic

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JGEC/B.TECH/CSE/PEC-(CS)-602B/2024-25

2025

Information Theory and Coding

Full Marks: 70 Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer ***all*** questions 5x2=10

1. Who invented BCH code? 2
2. Define Hamming Code. 2
3. State properties of Block codes. 2
4. Define Cyclic code. 2
5. For (7,4) Hamming code, if i=(1 0 1 0), find the codeword. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any ***four*** questions 4x15=60

6. a)Write and explain the algorithm of Hamming code.What do you understand by minimum Hamming distance?What is the hamming distance between the codes '11001011' and '10000111? 5+2+2

   b)Encode a binary word 11001 into the even parity hamming code. 6

7. a) Write the properties of the cyclic code. ? Explain the error detection and correction procedure of cyclic code with an example. 4+5

   b)The generator polynomial of a (7,4) cyclic code $G(x)=x^3+x+1$.Obtain all the code vectors for systamatic and non systamatic form. 6

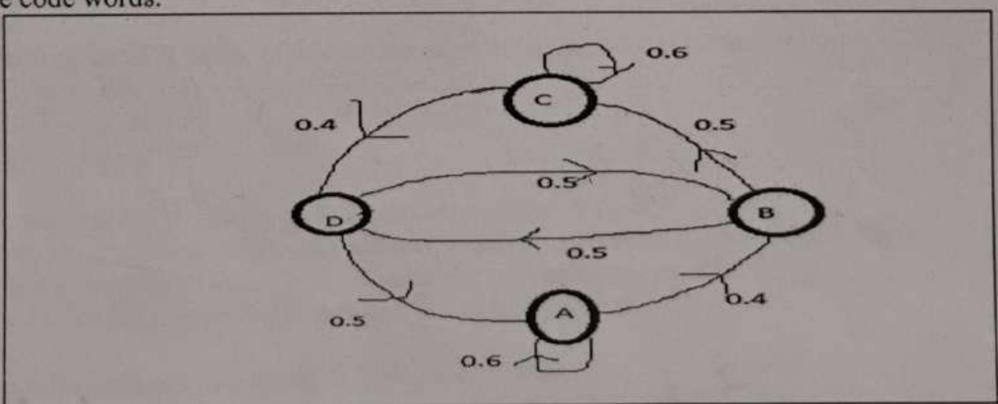
8. a) What is linear block code? What is parity check matrix? 2+2+5

   b)Consider a (6,3) systematic linear block code whose parity check matrix is given by
   $$H=\begin{bmatrix}1 & 1 & 0 & 1 & 0 & 0\\ 1 & 0 & 1 & 0 & 1 & 0\\ 0 & 1 & 1 & 0 & 0 & 1\end{bmatrix}$$
   determine the syndrome S when received codeword is R=111000 and R=111111

   b) The generator matrix of a (6,3) block code is $G=\begin{bmatrix}1 & 0 & 0 & 0 & 1 & 1\\ 0 & 1 & 0 & 1 & 0 & 1\\ 0 & 0 & 1 & 1 & 1 & 0\end{bmatrix}$. 6

   Find all the code words.

9. (i) 4+3+2



   Consider the Mark-off source given & compute the following
   (a)state probabilities.
   (b)Entropy of each state
   (c) Entropy of the source.

   (ii)Briefly explain channel capacities of different types of channels. 6

10. (i)Define entropy. Consider a discrete memoryless source with a source alphabet A = {so, s1, s2} with respective probs. $p_0 = ¼$, $p_1 = ¼$, $p_2 = ½$. Find the entropy of the source 2+4

(ii)A source emits one of the probable messages with probabilities 7/16, 5/16, 1/8 & 1/8 respectively. Prove the 2nd Extension property for the given data set. 4

(iii) State the properties of Entropy. 5

11. (i)Using Shannon-Fano algorithm, find Entropy, efficiency for six messages occurring with probabilities, 1/24, 1/12, 1/24, 1/6, 1/3, 1/3. 4+3

(ii) Taking the same data set design Huffman coding Theorem & determine Entropy & efficiency. 5+3

8

4.60

$\frac{1}{12} \cancel{\frac{2}{24}} \log_2 24$

$\frac{2}{3} \log_2 3$ $\frac{1}{6} \log_2 6$

$\frac{1}{12} \log_2 12$

$\frac{1}{2}$, $\frac{1}{3}$ + $\frac{2}{6}$ $\frac{1}{3}$ $\frac{1}{6}$ $\frac{2}{24}$ $\frac{1}{12}$

$\frac{2}{3}$

$\frac{1}{3}$, $\frac{2}{3}$

$\frac{1}{3}$, $\frac{1}{6}$ + $\frac{2}{12}$ $\frac{1}{6}$ $\frac{2}{24}$ $\frac{1}{12}$

$\frac{2}{6}$ $\frac{1}{3}$

$\frac{1}{6}$ + $\frac{1}{12}$ $\frac{1}{3}$

$\frac{2+1}{12} = \frac{3}{12}$ $\frac{1}{4}$

$\frac{1}{12} + \frac{1}{12} \Rightarrow \frac{2}{12}$ $\frac{1}{6}$

2^4 2 3

$\frac{1}{12}$ $\frac{1}{6}$ $\frac{2}{3}$

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2025

COMPILER DESIGN

Full Marks: 70 Times: 3 Hours

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Red Load/store
Flow control
Dead code
$A \to A\alpha \mid \beta$
$A \to \beta A'$
$A' \to \alpha A' \mid \epsilon$

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

null sta elim
Const folding
Strength reduction

Answer *all* questions

1. What is left recursion and with an example show how to eliminate it.
2. What is the difference between LL(1) and LR(1) parsers.
3. Define an ambiguous grammar. Give an example.
4. What is viable prefix? Give an example.
5. What do you mean by strength reduction in code optimization? Give example

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15 = 60

Answer any ***four*** questions

6. i) Briefly describe the structure of a compiler with suitable examples. 7
   ii) Write an unambiguous grammar which generates valid arithmetic expressions containing only 3 operators, exponentiation (^), multiplication(*) and addition(+) .The operators are right, left and left associative respectively and are mentioned in the order of higher to lower precedence level. 6
   iii) Write a regular expression for whitespace. 2

7. i) Write an algorithm to construct a predictive parsing table. Generate a predictive parsing table for the following grammar. 4+7

$$S \to aBh \mid Ce$$
$$B \to cC$$
$$C \to bd \mid \varepsilon$$

   ii) Generate the parse tree for the sentence, "acbdh" using the parsing table derived from 7i) clearly showing the moves made by the predictive parser. 4

8. Construct an SLR parsing table for the expression grammar given below. 15

$$E \to E + T \mid T$$
$$T \to T \star F \mid F$$
$$F \to (E) \mid id$$

9. i) Find out the three-address code for the expression x = (a+b)*(-c)/d 3+6
   Write the quadruple, Triple and Indirect triple representation for the above expression
   ii) Explain Loop Unrolling and Loop Jamming with suitable example. 5
   iii) Loop Optimization is Machine dependent or Independent 1

10. i) Write an algorithm to construct an SLR parsing table. 7
    ii) Differentiate among 'lexeme', 'pattern' and 'token with suitable examples. 3
    iii) Draw a transition diagram for unsigned numbers. 5

11. i) Construct a Syntax tree and DAG for the expression a+ a*(b-c)+ (b-c)*d. 2+2
    ii) Discuss in detail about various peephole optimization techniques 6
    iii) Explain synthesized and inherited attributes with example .What is basic block? 2+2+1

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SOFT SKILLS AND INTERPERSONAL COMMUNICATION

Full Marks: 70 Time: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions: 5x2=10

1. What attributes are regarded as soft skills? 2
2. Who coined the term "know thyself"? How can one know oneself? 2
3. When does attitude change? ABC 2
4. How is listening different from hearing? 2
5. What do you mean by effective communication? 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions: 4x15= 60

6. What is SWOT analysis? What are the benefits of SWOT analysis? 15
7. What do you mean by attitude? What are the important features of attitude? 15
8. What are the three basic listening modes? 15
9. What are the different barriers to communication? Explain. 15
10. What is a team? What are the aspects of team building? What skills are needed for teamwork? 3+5+7
11. What are the roles of a team leader? 15
12. What are the skills required in a group discussion? 15

Credits 2 → E

201/21

9.476 199/21